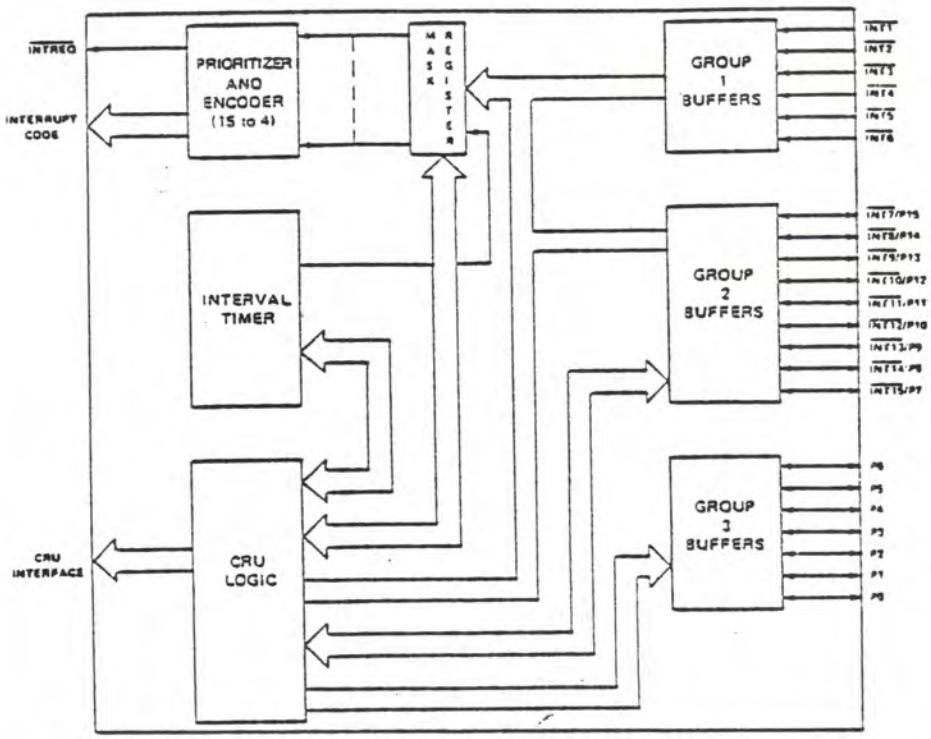


Figure 5. TMS 9901 Architecture

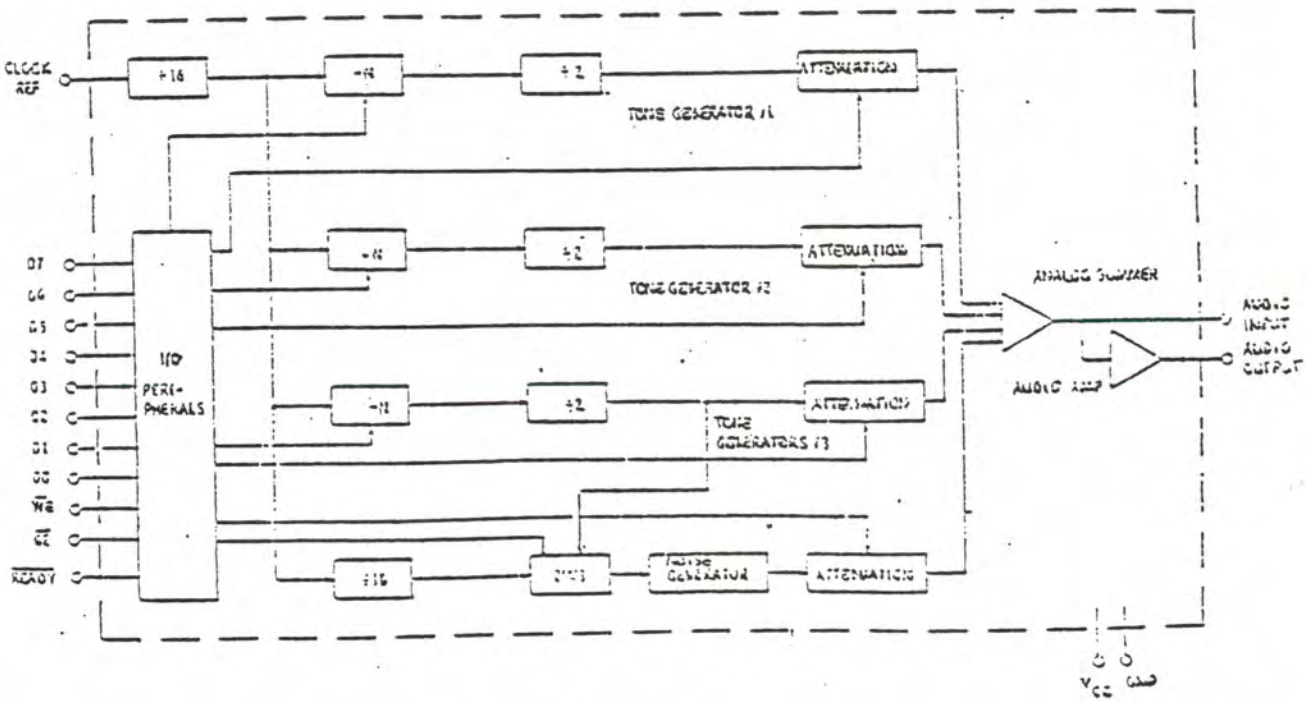


Figure 6. TIM 9919 Architecture

motor control on the cassette peripheral. The MAG OUT circuit takes the output from a programmable I/O line on the 9901 across a resistor-capacitor filtering circuit directly to the MAG OUT line. The MAG IN line uses two 4558 operational amplifiers. The first op-amp is used as an amplifier with a negative feedback network while the second takes the first's signal and modifies it into a usable TTL compatible input signal to the 9901.

The Mechanical Joystick Interface and the 40-key keyboard both make use of a 2-line-to-4-line decoder/demultiplexer (74LS156) in their interface with the TMS 9901. The joystick interface utilizes 5 Interrupt lines, 4 dedicated and 1 programmable, and two dedicated I/O lines. The Interrupt lines receive data from the joysticks by means of loading down the normally high lines. The two I/O lines are used to send data to the 2-line-to-4-line demultiplexer by enabling the joysticks. By enabling the joysticks, data is inputted through the interface to the 9901.

The 40-key keyboard uses three dedicated I/O lines and eight Interrupt lines, 4 dedicated and 4 programmable. The keyboard is divided into two 20-key sections, each 4 rows by 5 columns. The row data is sent directly to the Interrupt lines while the column data goes through the demultiplexers to the I/O lines. All of the keyboard lines are normally high.

SOUND GENERATION:

Another of the TI 99/4's key features is sound generation.

The key to music and sound generation within the system is TIM 9919 (SN 76489) Sound Generation Controller. The 9919 provides sound in three tones and over four octaves. Some of the features within the sound chip are listed:

* Three Programmable Tone Generators
* Programmable White Noise Generator
* Programmable Attenuation
* TTL Compatible
* 3.579 MHz Reference Clock
* Allows Simultaneous Sounds
* 8-ohm Speaker Drive Capability
* Single +5 volt Power Supply
* Packaged as 16-pin DIP

The TIM 9919 is a TTL compatible I^2L/Bipolar IC. Designed especially for use in microprocessor systems, it comes in a 16-pin DIP. The sound chip has 8 pins for input data bus lines. It also has 2 pins for its power requirements, one for +5v supply and one for ground reference. The remaining five signals are used for control and timing signals plus audio I/O. The TIM 9919 requires a 3.59MHz clock for a reference frequency in its sound generation. The 9900 also supplies three control signals ($\overline{CE}$, READY, and DBIN) to the sound chip. The remaining two pins are used for Audio Signal In and Audio Drive Out.

The TIM 9919 is made up of three programmable tone generators, a programmable noise generator, a clock scalor, individual generator attenuators and an audio summer output buffer. The TIM 9919's architecture can be seen in Figure 6. The three

programmable tone generators are composed of a frequency synthesis section and an attenuation section. The tone generators require two data words to operate. The frequency synthesis section uses a 10-bit input data word to produce an output signal of the desired frequency. This desired frequency is then inputted into a four-stage attenuator along with the four remaining input data bits to provide a maximum attenuation of 28 dB.

The noise generator is also composed of two sections, a noise source and an attenuator. The noise source is made of a shift register utilizing an exclusive OR feedback system and resulting in either Periodic or White noise generation. This output is also inputted to an attenuator to provide a maximum of 28 dB attenuation.

The audio summer/output buffer is an operational amplifier summing circuit. This circuit sums up the three tone generator outputs, the noise generator output, and the Audio Signal In. This allows for simultaneous sound. The output buffer can then send this signal out at up to 100 mA into an 8-ohm speaker load.

The sound chip receives three control signals from the CPU. These are the $\overline{CE}$ (Chip Select), $\overline{WE}$ (Write Enable), and READY signals. The $\overline{WE}$ signal when active indicates that the data bus has information from the 9900 available. The READY signal is activated when the sound chip has completed reading the data. $\overline{CE}$ is activated when data is able to be transferred from the 9900 to the sound chip.

REFERENCES:

1. 9900 Family Systems Design And Data Book; Texas Instruments, Inc. 1978.

2. The TTL Data Book for Design Engineering; Texas Instruments, Inc. 1976.

3. Microprocessor Data Manual; Dave Bursky - Electronic Design, 1978.

4. Microprocessor Interfacing Techniques; Austin Lesea and Rodney Zaks, 1978.

5. Digital Design with Standard MSI and LSI; Thomas R. Blakeslee, 1975.

6. Advanced Circuits - SN76489 Sound Generation Controller; Texas Instruments, Inc. 1978.

7. Electronic News; John Crudde, June 4, 1979.

8. Advanced Circuits - TMS 9918 Video Display Processor, Texas Instruments, Inc. 1978.

T I 9 9 / 4 T E S T S E T E R R O R T A B L E

FAILURE #	ERROR MSG	ERROR PARAMETER	ALLOWED VALUES HIGH	LOW	TABLE #
0	SYSTEM VOLTAGE	GROM +5 V	1075	972	1
		I/O +5 V	1075	972	2
		VIDEO +12 V	1283	1049	3
		GROM -5 V	- 972	- 1075	4
		I/O -5 V	- 972	- 1075	5
		GROM GROUND	- 61	- 225	6
1	SYSTEM CLOCK	I/O 3.0 MHZ	30009	29991	7
		GROM CLOCK	4800	4400	8
12	CASS. MOTOR CONTROLS	CS1 ON	350	0	9
		CS1 OFF	1076	972	10
		CS2 ON	358	0	11
		CS2 OFF	1076	972	12
13	CASS. OUTPUT	OUTPUT ON	1116	372	13
15	AUDIO GATE	OFF (VIDEO)	10	0	14
		ON (VIDEO)	310	290	15
18	9919 TONE BAD	9919 REG 0 VID	115	105	16
		9919 REG 1 VID	115	105	17
		9919 REG 2 VID	115	105	18
		9919 REG 0 EAR	115	105	19
		9919 REG 1 EAR	115	105	20
		9919 REG 2 EAR	115	105	21
19	9919 TONE BAD	9919 REG 0 VID	14683	13285	22
		9919 REG 1 VID	14683	13285	23
		9919 REG 2 VID	14683	13285	24
20	TONE ATTEN.	REG 0 0 DB	1590	530	25
		REG 1 0 DB	1590	530	26
		REG 2 0 DB	1590	530	27
21	TONE ATTEN.	REG 0 - 2 DB	1044	348	28
		REG 1 - 2 DB	1044	348	29
		REG 2 - 2 DB	1044	348	30
22	TONE ATTEN.	REG 0 - 4 DB	699	233	31
		REG 1 - 4 DB	699	233	32
		REG 2 - 4 DB	699	233	33
23	TONE ATTEN.	REG 0 - 8 DB	400	119	34
		REG 1 - 8 DB	400	119	35
		REG 2 - 8 DB	400	119	36

FAILURE #	ERROR MSG	ERROR PARAMETER	ALLOWED VALUES HIGH	LOW	TABLE #
24	TONE ATTEN.	REG 0 -16 DB	181	61	37
		REG 1 -16 DB	181	61	38
		REG 2 -16 DB	181	61	39
25	TONE ATTEN.	REG 0 OFF	25	-25	40
		REG 1 OFF	25	-25	41
		REG 2 OFF	25	-25	42
26	NOISE ATTEN.	WHITE 0 DB	2443	815	43
		WHITE - 2 DB	2392	798	44
		WHITE - 4 DB	1924	642	45
		WHITE - 8 DB	1222	408	46
		WHITE -16 DB	412	138	47
		WHITE OFF	25	-25	48
		PERIOD 0 DB	2443	815	49
		PERIOD - 2 DB	2392	798	50
		PERIOD - 4 DB	1924	642	51
		PERIOD - 8 DB	1222	408	52
		PERIOD -16 DB	412	138	53
		PERIOD OFF	25	-25	54
27	EXTERNAL AUDIO	I/O PORT	1253	1133	55
		CASSETTE PORT	1253	1133	56
28	AUDIO SUMMING	1 REG ON	160	54	57
		2 REG ON	402	134	58
		3 REG ON	847	283	59
		4 REG ON	1176	392	60
		4 REG + EXT	1882	628	61
--	UUT NOISE HIGH	TESTER CALIB	410	---	--

NOTE : ON AN ERROR, THE TEST SET WILL ABORT THE REMAINING MEASUREMENTS FOR THE TEST CURRENTLY IN PROGRESS, WITH THE UNMEASURED PARAMETERS LISTED WITH A 0 VALUE.

THE TWO NUMBERS LISTED AFTER AN ERROR PRINTOUT INDICATE THE TABLE # AND THE ACTUAL PARAMETER MEASUREMENT OF THE LAST ADC OR COUNTER CYCLE. THESE MAY OR MAY NOT BE ASSOCIATED WITH THE ERROR.

(.PAULH.HOMF.PARLST)

15	AUDIO GATE	A CASSETTE INPUT IS GENERATED AS ABOVE AND THE AUDIO GATE IS TURNED ON AND OFF TO VERIFY CORRECT OPERATION. A FAILURE ON THIS TEST ALONG WITH TEST 14 PROBABLY MEANS THE SIGNAL WAS NOT GETTING TO TRANSISTOR Q405. A FAILURE OF THIS TEST ALONE INDICATES Q405 OR ITS COMPANION COMPONENTS MAY BE DEFECTIVE OR THE SOUND CHIP (U511) IS BAD. THE MEASURED PARAMETER IS THE FREQUENCY OF THE INPUT SIGNAL AND SHOULD BE 300 HZ.
16	GROM RESET	THE RESET LINE ON THE GROM PORT WAS PULLED TO -5 V AND THE UUT SHOULD HAVE RESET. AN ERROR MEANS A FAILURE IN THE CIRCUITRY ASSOCIATED WITH PIN 1 OF THE GROM CONNECTOR. ALSO, A FAILURE ON THIS TEST MAY CASE THE REMAINDER OF THE TESTS TO FAIL, SINCE THE UUT WOULD NOT BE RUNNING THE GPL TEST SOFTWARE. PIN 21 OF THE GROM CONNECTOR MAY ALSO CAUSE THIS ERROR.
17	I/O INTERRUPT	THE UUT DID NOT ACKNOWLEDGE A TEST SET GENERATED INTERRUPT. SUSPECT H/W INCLUDES : I/O PORT PIN 4 EXT INT TMS 9901 U300
18	9919 TONE BAD	THE UUT EXECUTES GPL SOFTWARE TO EXERCISE THE TMS 9919 SOUND CHIP. THIS TEST CHECKS A LOW AND HIGH FREQUENCY ON EACH TONE REGISTER, USING THE TEST SET'S FREQ. COUNTER. AN ERROR MAY BE CAUSED BY A DEFECTIVE SOUND CHIP, OR AN OPEN ON EITHER THE EARPHONE CONNECTOR OR THE VIDEO CONNECTOR. A FAILURE HERE WILL CAUSE THE REST OF THE SOUND TESTS TO BE SKIPPED.
19-25	TONE ATTEN.	THE UUT RUNS THRU A SERIES OF ATTENUATION TESTS WITH THE 9919 TONE REGISTERS. THESE VALUES ARE MEASURED USING THE TEST SET ADC, BUT THE VALUES ARE RELATIVE ONLY AND CANNOT BE STATED AS AN ABSOLUTE VOLTAGE. A FAILURE ON THIS TEST ALONE MAY NOT NECESSARILY MEAN A BAD SOUND CHIP, DEPENDING ON THE DEVIATION FROM THE ALLOWED VALUES.
26	NOISE ATTEN.	THE SAME AS THE PREVIOUS TEST BUT RUN ON THE TWO NOISE REGISTERS IN THE SOUND CHIP. USE THE SAME DISCRETION AS EXPLAINED ABOVE IN FAILING A UNIT BASED ON A FAILURE HERE ONLY.
27	EXTERNAL AUDIO	THE TEST SET GENERATES A 1200 HZ TONE AND APPLIES IT TO THE TWO AUDIO INPUTS. A FAILURE HERE, ALONG WITH A FAILURE OF THE AUDIO GATE, USUALLY INDICATES A BAD TMS9919 (U511). IF THE CASSETTE PORT INPUT FAILS BUT THE I/O PORT INPUT PASSES, THEN THE ERROR IS IN THE CASS. AUDIO CIRCUITRY. THE TWO INPUTS ARE : I/O PORT PIN 44 SOUND IN CASS. PORT PIN 4 CASS. AUDIO IN THE OUTPUT IS VERIFIED BY THE TEST SET'S FREQ COUNTER LOOKING AT THE EARPHONE CONNECTOR.

28	AUDIO SUMMING	THE UUT TURNS ON SUCCESSIVE TONE REGISTERS AND THE TEST SET VERIFIES THE SOUND AMPLITUDE OUT OF THE SOUND CHIP USING THE ADC. THE VALUES MEASURED ARE RELATIVE VOLTAGES ONLY AND ARE NOT TO BE RELATED TO AN ABSOLUTE LEVEL. A FAILURE HERE MEANS THE SOUND CHIP (U511) IS FAULTY OR THE EARPHONE CONNECTION IS NOT FUNC-TIONING.
***	OTHER ERRORS	THERE ARE THREE ADDITIONAL ERRORS THAT MAY OCCUR DURING THE TESTING CYCLE. ONLY ONE, THE `ABORT`, MAY INDICATE A UUT FAILURE. THESE THREE ARE :
	* ABORT *	THIS MEANS THE OPERATOR ABORTED TESTING WHILE THE TEST CYCLE WAS STILL IN PROGRESS. THERE SHOULD BE A NOTE FROM THE OPERATOR EXPLAINING THE REASON FOR THIS ACTION.
	TESTER GROMS	WHEN THIS MESSAGE APPEARS, DURING THE GROM BYTE-BY-BYTE COMPARISON TEST (STEP #7) THE INTERNAL REFERENCE GROMS IN THE TEST SET FAILED TO PERFORM CORRECTLY. R & M SHOULD BE NOTIFIED WHEN THIS ERROR OCCURS.
	UUT NOISE HIGH	EVERY TIME THE TEST SET PERFORMS A LEVEL CHECK ON THE EARPHONE OR THE CASSETTE MAO OUT PORT, AN ADDITION CHECK IS MADE ON THE NOIS LEVEL AT THESE CONNECTORS. THIS MESSAGE INDICATES THE MEASURED VALUE IS OUT OF TOLERANCE, WHICH MAY MEAN THE TEST SET NEEDS TO BE RECALIBRATED. REPORT THE OCCURANCE TO R & M.